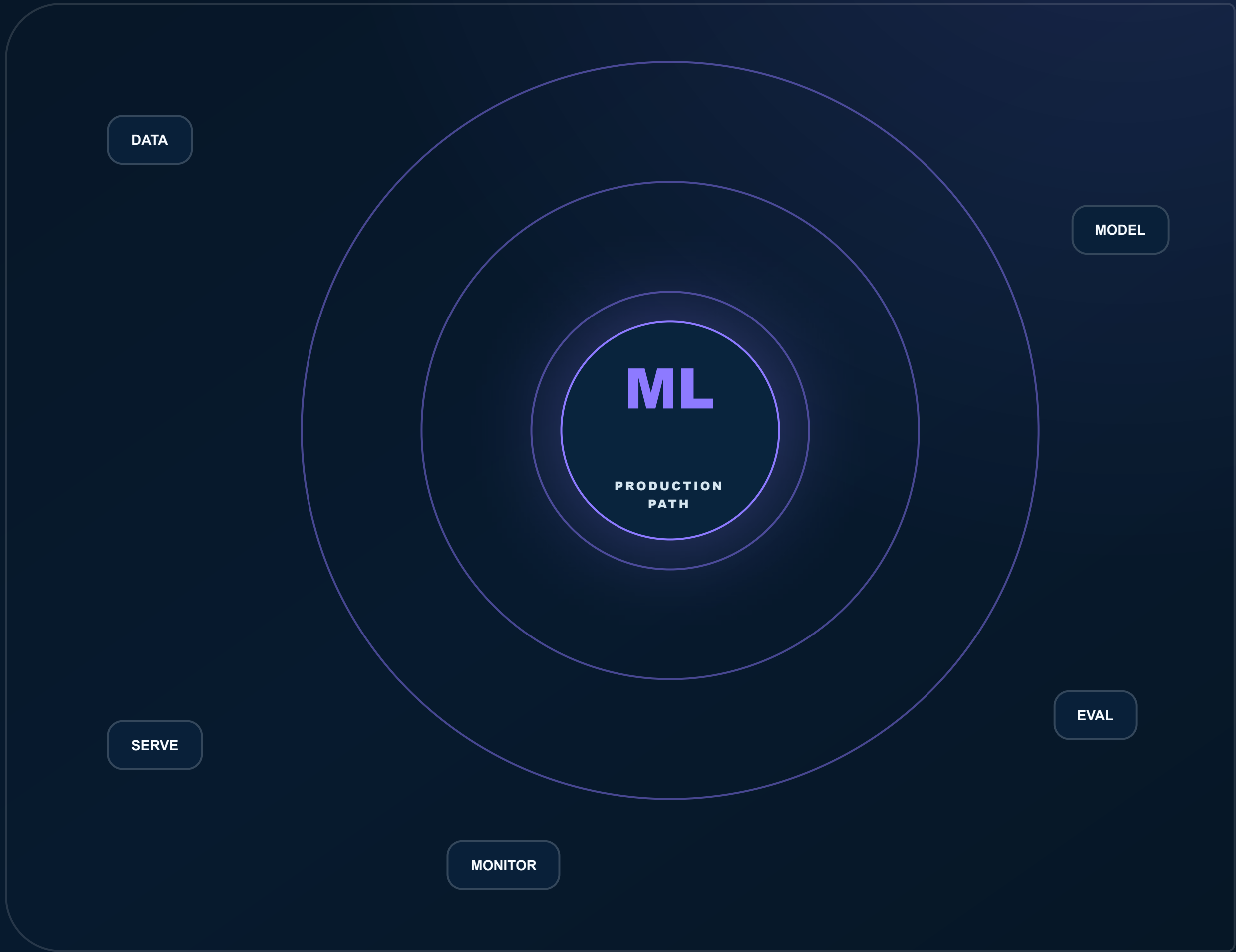
Artificial Intelligence · Multi-level

From data foundations to production ML.

A production-oriented machine learning path spanning classical ML, deep learning, computer vision, NLP, voice, multimodal AI, model adaptation and MLOps.

Hybrid guided learning | 20-28 weeks | 6 modules

Explore the program →



Build and evaluate classical ML
Build and evaluate classical ML systems

Develop deep learning solutions for
Develop deep learning solutions for CV and NLP

Fine-tune models with PEFT, LoRA
Fine-tune models with PEFT, LoRA and QLoRA

Train a small educational transformer
Train a small educational transformer

THE SKILLS THE ROLE NEEDS

Applied Machine Learning Engineer Path - learned by building, not memorising.

Data -> ML -> Deep Learning -> CV/NLP -> LLM/SLM Engineering -> MLOps. Every capability is connected to deliberate practice, evidence and review.

01

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05

Python and Pandas

Python and Pandas

06

scikit-learn

scikit-learn

The learning journey

Move from understanding to demonstrable capability.

1 Understand → 2 Practice → 3 Build → 4 Review → 5 Demonstrate

CORE PROGRAM

From foundation to applied delivery.

20-28 weeks · 6 current BRD-aligned modules · practice and evidence in every stage.

01 Data, Math and Python Bridge

Practical foundations for ML work.

Python for data · Statistics essentials · Data preparation · SQL and visualization

02 Classical Machine Learning

Supervised, unsupervised and evaluation workflows.

Regression and classification · Feature engineering · Clustering · Metrics and model selection

03 Deep Learning

Neural networks and practical training.

Neural network foundations · Optimization · CNNs · Sequence and transformer foundations

04 Computer Vision, NLP and Multimodal

Modern perception and language systems.

Classification and detection · Segmentation, tracking and OCR · NLP and transformers · Voice and multimodal workflows

05 LLM and SLM Engineering

Adapt, compress and understand language models.

Dataset preparation and evaluation · Supervised fine-tuning · PEFT, LoRA and QLoRA · Quantization and distillation

06 Educational Transformer and MLOps

Understand internals and operate models.

Tokenizer and embeddings · Transformer blocks and training loop · Model serving · Monitoring, drift and capstone

TOOLS & PRACTICE

Build with the current working set.

- ML Python and Pandas
- ML scikit-learn
- ML PyTorch
- ML OpenCV and YOLO concepts
- ML Transformers and PEFT
- ML Model serving and MLOps

CAPSTONE DIRECTION

Frame a realistic problem, build the role-appropriate artifact, explain the trade-offs, respond to review and complete the published criteria.

OUTCOMES & BEYOND

Build skill. Build confidence. Build evidence.

No inflated promises - a clear route from learning to reviewable work and verified completion.

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Ideal participants

1. AI and data students
2. Data analysts
3. Software developers entering ML
4. Career switchers

What you will demonstrate

01 · Build and evaluate classical ML systems

Documented, reviewable and connected to the learning record.

02 · Develop deep learning solutions for CV and NLP

Documented, reviewable and connected to the learning record.

03 · Fine-tune models with PEFT, LoRA and QLoRA

Documented, reviewable and connected to the learning record.

04 · Train a small educational transformer

Documented, reviewable and connected to the learning record.

START YOUR JOURNEY

Applied Machine Learning Engineer Path

Hybrid guided learning. Current schedule, fees, previews and seat availability appear before checkout.

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